

EDUCATION

University of Michigan <i>B.S.E. in Robotics, GPA: 4.0/4.0</i>	Ann Arbor, MI <i>Anticipated December 2026</i>
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Graduate-Level Coursework:
Foundation of Language Models, Mobile Robotics: Methods and Algorithms, Multi-Robot Systems

Shanghai Jiao Tong University <i>B.S.E. in Electrical and Computer Engineering</i>	Shanghai, China <i>Anticipated 2027</i>
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RESEARCH EXPERIENCE

University of Michigan Scalable Spatial Intelligence Laboratory <i>Undergraduate Researcher</i>	Ann Arbor, MI <i>February 2026-Present</i>
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- **Constructed** an open-vocabulary multi-robot spatial representation and coordination system in Isaac Sim.
- **Exploring** zero-shot semantic capability-task binding coordination across heterogeneous multi-robot teams.

University of Michigan Computational Autonomy and Robotics Laboratory <i>Undergraduate Researcher</i>	Ann Arbor, MI <i>October 2025-Present</i>
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- **Designed** a diagnostic benchmark to evaluate VLMs' sensorimotor mapping capability and **investigated** the bottleneck of detailed information loss within the VLM pipeline.
- **Assisted** in developing Vision-and-Language Navigation (VLN) framework, **exploring** scene graph generation, supervised fine-tuning, and semantic world model's idea for VLN.

Shanghai Jiao Tong University Flow Diagnostics and Analysis Lab <i>Undergraduate Researcher</i>	Shanghai, China <i>April 2024-August 2025</i>
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- **Engineered** a versatile, high-fidelity experimental platform from scratch, **creating** a modular apparatus capable of accommodating diverse fluid-structure interaction scenarios; **accepted** as the lab's standard setup.
- **Integrated** a multi-modal data acquisition pipeline fusing high-speed vision with piezoelectric sensors, **enabling** synchronized morphological and force measurements for micro-impact analysis.
- **Corroborated** experimental measurements with theoretical impact models, **conducting** consistency analysis to validate the platform's fidelity and prototyping preliminary data-driven baselines.

Tsinghua University Shenzhen Graduate School <i>Research Intern</i>	Shenzhen, China <i>December 2023 - February 2024</i>
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- **Identified** critical bottlenecks in edge-deployed active perception through a comparative analysis of current UAV path planning paradigms, **contributing** to a hybrid exploration algorithm that optimizes the exploration-exploitation trade-off.
- **Assisted** in the sim-to-real transfer, **validating** the planning algorithm on physical UAV platforms to assess real-world performance.

WORK EXPERIENCE

Edan Instruments, Inc. <i>Research & Development System Intern</i>	Shenzhen, China <i>December 2024 - February 2025</i>
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- **Prototyped** the company's first interactive 3D visualization system for electrocardiographic diagnostics, **securing** management approval for production integration.
- **Assisted** in refining diagnostic algorithms.

Global College, Shanghai Jiao Tong University

Tech Teaching Assistance for Introduction to Engineering

Shanghai, China

September 2024 - December 2024

- **Mentored** freshman engineering teams through the full mechatronics design cycle, **delivering** hands-on technical instruction in C++, SolidWorks and Arduino.
- **Co-designed** innovative projects, **Nominated** for Annual Outstanding TA award.

LEADERSHIP EXPERIENCE

Shanghai Jiao Tong University Global College Tennis Association

Shanghai, China

Founding President

August 2024 - August 2025

- **Established** democratic governance structure by organizing founding elections and drafting comprehensive club bylaws, **creating** sustainable organizational framework for 100+ members.
- **Coordinated** regular training sessions and skill development programs to improve member performance and engagement, **leading** institute team participation in inter-university competitions.
- **Developed** digital platform connecting members for practice partnerships.

SKILLS

- **Programming Languages:** C++, Python, MATLAB, C, Verilog
- **Robotics:** ROS 2, VLM, SLAM, Isaac Sim, Habitat
- **Tools:** Git/Linux, Docker, LaTeX, SolidWorks, High-speed Imaging
- **Language:** Chinese (Native) and English (Proficient)

REFERENCES

Dr. Yulun Tian

Scalable Spatial Intelligence Laboratory Principal Investigator

Assistant Professor of Robotics

February 2026 - Present

University of Michigan

Email: yulunt@umich.edu

Dr. Maani Ghaffari

Computational Autonomy and Robotics Laboratory Principal Investigator

Assistant Professor of Robotics

October 2025 - Present

University of Michigan

Email: maanigj@umich.edu

Dr. Yue Hu

Postdoctoral Fellows

October 2025 - Present

University of Michigan

Email: huyu@umich.edu

Dr. David L.S. Hung

Flow Diagnostics and Analysis Lab Principal Investigator

Full Professor and Associate Dean of Shanghai Jiao Tong University Global College

April 2024 - August 2025

Shanghai Jiao Tong University

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Dr. Xinlei Chen

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